Aenderungsbeschreibung/Description of Revision	Kommt vor/Frequency
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Changes from Var1 Ed 01.04 to Var1 Ed 01.06

Corrections:

Scaling Analog Reg 209 -> P-Exhaust Diff SCR1 to 10000
Scaling Analog Reg 1198+1199 -> LM HI/HIHI P-Exhaust Diff SCR1 to 10000
Renaming Alarm Reg 4000-4078
Min or Max Range changed for Analog Reg.: 6,7, 24-27, 30-39, 43, 46-49, 56-57, 70, 75-76, 78-88, 122-133, 141, 150, 170-173, 183

New measuring points added:

Analog Register 16 RatedPower
Analog Register 17 MaxEnginePower

New limits added:

LM LO P-Gas Rail Bank A Register 1207
LM LOLO P-Gas Rail Bank A Register 1208
LM LO T-Gas Register 1209
LM LOLO T-Gas Register 1210
LM HI P-Gas Rail Bank A Register 1211
LM HIHI P-Gas Rail Bank A Register 1212
LM HI P-Gas Rail Bank B Register 1215
LM HIHI P-Gas Rail Bank B Register 1216
LM HI T-Gas Register 1217
LM HIHI T-Gas Register 1218
LM HI T-Coolant Intercooler Register 1219
LM HIHI T-Coolant Intercooler Register 1220

New alarms added:

MG Cold Start required Register 1910
AL MMS Internal Com. Lost Register 4946
HI NOx Wert Register 4948
AL Emission Fault Register 4950
SD P-Gas Inlet before Filter Register 4952
SD P-Gas Inlet after Filter Register 4954
SD P-Gas GRU Block-Bleed Register 4956
SD P-Gas GRU Outlet Register 4958
SD T-Gas GRU Outlet Register 4960
SD P-Control Air GRU IP-Conv Register 4962
SD P-Control Air GRU Inlet Register 4964
LOLO P-Gas GRU Inlet Register 4966
HIHI P-Gas GRU Inlet Register 4968
LO P-Gas Rail Register 4970
HI P-Gas Rail Register 4972

Tolerierung/Tolerancing ISO 8015 | Masse/Size ISO 14405
Allgemeintoleranzen/General Tolerances ISO 2768-mK

Masse/Mass kg | Format/Size **A4**

Bauart/Type: GenolineNG

Ausf. u.Lieferung Tech. Characteristics

Allgemeine Fertigungsvorschriften nach MMN332
Production Specification per MMN 332

Werkstoff/Material

Oberflaechenschutz Surface Protection

Oberflaechenangaben nach MTN5033
Surface Specification per MTN5033

Halbzeug,Modell Gesenk / Semifinished Product Pattern, Die

Verwendbar fuer Typ/Applicable to Model: GLNG Release 30

Projekt-/Auftrags-Nr./Project/Order No.

Fert.Ber. Man.Adv

Mont.Ber. Inst.Adv

Material-Nr./Material No.: XZ00E52000039

Ref.-Nr./ Ref. No.: Type1 Var01 ED01_04

Emission ID

WORD	Datum/Date	Name
Erstell. Drawn	30.03.2022	schaepers
Bearb. Change	31.03.2022	schaepers
Norm Std.	31.03.2022	ihmor stef
Gepr. Checked	31.03.2022	ihmor stef

Benennung/ Title /

DOKUMENTATION

GLNG Fremdschnittstelle Modbus RTU

DOCUMENTATION

GLNG Foreign MCS Interface Modbus RTU

A Rolls-Royce solution
Rolls-Royce Solutions GmbH

Zeichnungs-Nr./Drawing No. ZNG00045268

Beschreibung/Description Type1 Var01 ED01_06

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T-468-0107

MCS Analog Measuring Points

Function Code	03 Holding Register
	16 Write Holding Register
Minimum Cycle Time	200ms
Missing Data:	0x7FFF
Sensor Defect	0x8001
Sensor Defect Combined Registers (32Bit)	0x80000001

Register	Denomination	Min	Max	Unit	Scaling	Read/Write
0	Engine Speed	0	3000	rpm	10	R
1	Engine Speed SaSy	0	3000	rpm	10	R
2	Engine Speed Camshaft	0	3000	rpm	10	R
3	Engine Speed Crankshaft	0	3000	rpm	10	R
4	Idle Speed	0	3000	rpm	10	R
5	Nominal Speed	0	3000	rpm	10	R
6	Feedback Speed Demand	0	5000	rpm	10	R
7	Feedback Speed Demand Eff.	0	5000	rpm	10	R
11	ETC 1 Speed	0	100	krpm	100	R
12	ETC 2 Speed	0	100	krpm	100	R
13	ETC 3 Speed	0	100	krpm	100	R
14	ETC 4 Speed	0	100	krpm	100	R
16	Rated Power	0	10000	kW	100	R
17	Max Engine Power	0	10000	kW	100	R
24	Power Supply EIM/Starter Bat.	0	40	V	100	R
25	ECU Power Supply Voltage	0	40	V	100	R
26	U-SPU Main Power Supply	0	40	V	100	R
27	U-SPU Emerg. Power Supply	0	40	V	100	R
30	P-Lube Oil aft. Filter (ECU)	0	12	bar	100	R
31	P-Lube Oil aft. Filter SaSy	0	12	bar	100	R
32	P-Lube Oil A-Side ETCs	0	12	bar	100	R
33	P-Lube Oil B-Side ETCs	0	12	bar	100	R
34	P-Oil after Level Pump	0	12	bar	100	R
35	P-Lube Oil last bearing	0	12	bar	100	R
36	P-Piston Cooling Oil	0	12	bar	100	R
37	P-Piston Cool.Oil Filter Diff.	0	12	bar	100	R
38	P-Lube Oil before Filter (ECU)	0	12	bar	100	R
39	P-Oil Filter Difference	0	3	bar	100	R
42	P-Oil Refill Pump	0	6	bar	100	R
43	P-Coolant	0	10	bar	100	R
44	P-Raw Water before Pump	0	6	bar	100	R
45	P-Raw Water	0	6	bar	100	R
46	P-Fuel	0	16	bar	100	R
47	P-Fuel before Filter (ECU)	0	16	bar	100	R
48	P-Fuel Prefilter Diff.	0	2	bar	100	R
49	P-Fuel Filter Difference	0	2	bar	100	R
54	P-Crankcase	-70	70	mbar	100	R
55	P-Intake Air B	-70	70	mbar	100	R
56	P-Charge Air A	0	5	bar	100	R
57	P-Charge Air B	0	5	bar	100	R
58	P-Control Air	0	10	bar	100	R
59	P-Start Air	0	50	bar	100	R
60	P-Exhaust back	0	200	bar	100	R
62	UTC Seconds	0	59	digit	1	R
63	UTC Minutes	0	59	digit	1	R
64	UTC Hour	0	23	digit	1	R
65	Month	1	12	digit	1	R
66	Day	1	31	digit	1	R
67	Year	1970	2060	digit	1	R
70	T-Lube Oil	0	120	degC	10	R
71	T-Lube Oil ETC	-20	120	degC	10	R
75	T-Fuel bef.Low Press Pump	0	120	degC	10	R
76	T-Fuel Common Rail A	0	120	degC	10	R
77	T-Fuel Common Rail B	-20	120	degC	10	R
78	T-Coolant (ECU)	0	120	degC	10	R
79	T-Coolant SaSy	0	120	degC	10	R
80	T-Raw Water	0	120	degC	10	R
81	T-Intake Air	0	120	degC	10	R
82	T-Charge Air A	0	120	degC	10	R
83	T-Charge Air B	0	120	degC	10	R
84	T-Charge Air Seq Ctrl Valve	0	400	degC	10	R

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Register	Denomination	Min	Max	Unit	Scaling	Read/Write
85	T-Exhaust Combined A	0	1000	degC	10	R
86	T-Exhaust before ETC (Combined B)	0	1000	degC	10	R
87	T-Exhaust Combined C	0	1000	degC	10	R
88	T-Exhaust Combined D	0	1000	degC	10	R
89	T-Exhaust A1	0	1000	degC	10	R
90	T-Exhaust A2	0	1000	degC	10	R
91	T-Exhaust A3	0	1000	degC	10	R
92	T-Exhaust A4	0	1000	degC	10	R
93	T-Exhaust A5	0	1000	degC	10	R
94	T-Exhaust A6	0	1000	degC	10	R
95	T-Exhaust A7	0	1000	degC	10	R
96	T-Exhaust A8	0	1000	degC	10	R
97	T-Exhaust A9	0	1000	degC	10	R
98	T-Exhaust A10	0	1000	degC	10	R
99	T-Exhaust B1	0	1000	degC	10	R
100	T-Exhaust B2	0	1000	degC	10	R
101	T-Exhaust B3	0	1000	degC	10	R
102	T-Exhaust B4	0	1000	degC	10	R
103	T-Exhaust B5	0	1000	degC	10	R
104	T-Exhaust B6	0	1000	degC	10	R
105	T-Exhaust B7	0	1000	degC	10	R
106	T-Exhaust B8	0	1000	degC	10	R
107	T-Exhaust B9	0	1000	degC	10	R
108	T-Exhaust B10	0	1000	degC	10	R
109	T-Exhaust Mean	0	1000	degC	10	R
110	T-Bearing 1	0	200	degC	10	R
111	T-Bearing 2	0	200	degC	10	R
112	T-Bearing 3	0	200	degC	10	R
113	T-Bearing 4	0	200	degC	10	R
114	T-Bearing 5	0	200	degC	10	R
115	T-Bearing 6	0	200	degC	10	R
116	T-Bearing 7	0	200	degC	10	R
117	T-Bearing 8	0	200	degC	10	R
118	T-Bearing 9	0	200	degC	10	R
119	T-Bearing 10	0	200	degC	10	R
120	T-Bearing 11	0	200	degC	10	R
121	T-Bearing Mean	0	200	degC	10	R
122	T-Splash Oil B1	0	200	degC	10	R
123	T-Splash Oil B2	0	200	degC	10	R
124	T-Splash Oil B3	0	200	degC	10	R
125	T-Splash Oil B4	0	200	degC	10	R
126	T-Splash Oil B5	0	200	degC	10	R
127	T-Splash Oil B6	0	200	degC	10	R
128	T-Splash Oil B7	0	200	degC	10	R
129	T-Splash Oil B8	0	200	degC	10	R
130	T-Splash Oil B9	0	200	degC	10	R
131	T-Splash Oil B10	0	200	degC	10	R
132	T-Splash Oil Mean	0	200	degC	10	R
133	T-ECU	-40	120	degC	10	R
135	T-Exhaust after Engine	0	400	degC	10	R
140	ECU Failure Codes	0	5000	digit	1	R
141	Torque Limitation Code	-2147483646	2147483646	digit	1	R
142	SPU Interface Version HI	-32766	32766	digit	1	R
143	SPU Interface Version LO					
144	MCS Version HI	-2147483646	2147483646	digit	1	R
145	MCS Version LO					
146	SPU Alarm Groups coded static HI	-2147483646	2147483646	digit	1	R
147	SPU Alarm Groups coded static LO					
148	SPU Alarm Groups coded dyn. HI	-2147483646	2147483646	digit	1	R
149	SPU Alarm Groups coded dyn. LO					
150	Alarm Counter	-32766	32766	digit	1	R
152	Status Start Procedure HI	-2147483646	2147483646	digit	1	R
153	Status Start Procedure LO					
155	Trip Fuel Consumption HI	0	200000000	l	1	R
156	Trip Fuel Consumption LO					
157	Total Fuel Consumption HI	0	200000000	l	1	R
158	Total Fuel Consumption LO					
159	Actual Fuel Consumption	0	3000	l/h	10	R
160	Mean Trip Fuel Consumption	0	3000	l/h	10	R
161	Fuel Consumption Max	0	3000	l/h	10	R
162	Fuel Consumption Act	0	100	%	100	R
163	Fuel Consumption Mean Trip	0	100	%	100	R

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Register	Denomination	Min	Max	Unit	Scaling	Read/Write
165	Injection Limitation 0-120%	0	120	%	100	R
166	Injection in Relation to DBR	0	120	%	100	R
170	ECU Operat.Hours HI	0	999999	h	1	R
171	ECU Operat.Hours LO					
172	Trip Duration HI	0	999999	h	1	R
173	Trip Duration LO					
176	Engine Load Reserve	0	120	%	100	R
178	Injection Quantity 0-120	0	120	%	100	R
183	User Instruction	-32766	32766	digit	1	R
204	Cylinder Cutout Mask HI	-2147483646	2147483646	digit	1	R
205	Cylinder Cutout Mask LO					
206	T-Exhaust SCR Inlet1	0	1000	degC	10	R
207	T-Exhaust SCR Inlet2	0	1000	degC	10	R
208	T-Exhaust SCR Outlet1	0	1000	degC	10	R
209	P-Exhaust Diff SCR1	0	0,2	bar	10000	R
210	P-RA Filter Out At PFM	0	10	bar	100	R
211	T-Reducing Agent At PFM	0	1000	degC	10	R
212	RA Tank Level Norm	0	100	%	100	R
213	P-Fuel Prefilter	0	10	bar	100	R
220	T-Exhaust SCR Outlet 2	0	1000	degC	10	R
221	Power Actual Gen	0	2000	kW	10	R
231	T-Bearing GenDE	-50	200	degC	10	R
232	T-Bearing GenNDE	-50	200	degC	10	R
233	T-AirInlet Gen	-50	200	degC	10	R
234	T-AirOutlet Gen	-50	200	degC	10	R
236	T-WindingL1 Gen	-50	200	degC	10	R
237	T-WindingL2 Gen	-50	200	degC	10	R
238	T-WindingL3 Gen	-50	200	degC	10	R
239	T-WindingL4 Gen	-50	200	degC	10	R
246	T-Coolant Gen Out	-50	200	degC	10	R
247	T-Coolant Gen In	-50	200	degC	10	R
250	CPM Sensor Configuration 1	0	2147483646	digit	1	R
251	Cylinder Performance Cyl. A1	0	100	%	100	R
252	Cylinder Performance Cyl. A2	0	100	%	100	R
253	Cylinder Performance Cyl. A3	0	100	%	100	R
254	Cylinder Performance Cyl. A4	0	100	%	100	R
255	Cylinder Performance Cyl. A5	0	100	%	100	R
256	Cylinder Performance Cyl. A6	0	100	%	100	R
257	Cylinder Performance Cyl. A7	0	100	%	100	R
258	Cylinder Performance Cyl. A8	0	100	%	100	R
259	Cylinder Performance Cyl. A9	0	100	%	100	R
260	Cylinder Performance Cyl. A10	0	100	%	100	R
261	Cylinder Performance Cyl. B1	0	100	%	100	R
262	Cylinder Performance Cyl. B2	0	100	%	100	R
263	Cylinder Performance Cyl. B3	0	100	%	100	R
264	Cylinder Performance Cyl. B4	0	100	%	100	R
265	Cylinder Performance Cyl. B5	0	100	%	100	R
266	Cylinder Performance Cyl. B6	0	100	%	100	R
267	Cylinder Performance Cyl. B7	0	100	%	100	R
268	Cylinder Performance Cyl. B8	0	100	%	100	R
269	Cylinder Performance Cyl. B9	0	100	%	100	R
270	Cylinder Performance Cyl. B10	0	100	%	100	R
271	Cylinder Performance Mean	0	100	%	100	R
272	P-Gas A	0	1000	bar	100	R
273	T-Gas	0	1000	degC	10	R
274	Actual Gas Consumption	0	214748364	Kg/h	10	R
275	Trip Gas Consumption	0	214748364	t	10	R
276	Total Gas Consumption	0	214748364	t	10	R
277	GRU State	0	2147483646	digit	1	R
278	P-Gas Inlet	0	1000	bar	100	R
279	P-Gas Outlet	0	1000	bar	100	R
280	T-Gas GRU	0	1000	degC	10	R
281	P-Gas Set Point	0	1000	bar	100	R
282	Throttle Flap A Actual Value	0	100	%	100	R
283	Throttle Flap B Actual Value	0	100	%	100	R
284	P-Gas B	0	1000	bar	100	R
285	Mean Trip Gas Consumption	0	214748364	Kg/h	10	R
286	Max. Scale Gas Consumption	0	214748364	Kg/h	10	R
287	Actual Gas Cons. Bargraph	0	100	%	100	R
288	Mean Trip Gas Cons. Bargraph	0	100	%	100	R
289	T-EMU	0	100	degC	10	R
290	EMU Power Supply Voltage	0	50	V	10	R

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Register	Denomination	Min	Max	Unit	Scaling	Read/Write
291	T-CPM	0	100	degC	10	R
292	CPM Power Supply Voltage	0	50	V	10	R
293	T-Coolant Intercooler	0	1000	degC	10	R
294	T-EIM	0	100	degC	10	R
295	MMS Errorcodes	0	2147483646	digit	1	R
299	RA Concentration	0	100	%	100	R
300	P-Exhaust SCR Inlet1	0	2	bar	10000	R
301	P-Exhaust SCR Inlet2	0	2	bar	10000	R
302	P-Exhaust SCR Inlet 1 redundant	0	2	bar	10000	R
303	P-Exhaust Diff SCR2	0	0,2	bar	10000	R
305	P-Dosing Air SUDU 1	0	10	bar	100	R
306	SCR System State1	0	2147483646	digit	1	R
307	SCR System State2	0	2147483646	digit	1	R
308	Actual RA Consumption	0	100	l/h	10	R
309	TripRAConsumption HI	0	200000000	l	1	R
310	TripRAConsumption LO					
311	TotalRAConsumption HI	0	200000000	l	1	R
312	TotalRAConsumption LO					

MCS Alarms

Function Code		01 Read Coils
Minimum Cycle Time for sychronisation and initialisation		5000ms
Analog-Register 150 (Alarm Counter) is being changed if new Alarm occurs or Alarm disappears. If Register-Value has changed since last request, read all Alarm-Informations at once.		
2 Bits per Alarm		
High-Bit	Low-Bit	
0	0	No Alarm
0	1	Acknowledged Alarm
1	0	Spare/not used
1	1	Unacknowledged Alarm

Register	Denomination	Alarm Priority	Alarm Description
1002	LOLO Engine Speed	Yellow (3)	Engine is being stalled. The engine speed of the normally operating engine dropped below the limit from parameter 2.2500.027 Limit Engine Speed Low without any stop request. For safety reason the engine is stopped when this event occurs.
1004	HIHI Engine Overspeed (EIM)	Red (2)	Highest Limit of engine overspeed
1006	HIHI Engine Overspeed SaSy	Red (2)	An engine overspeed was detected by the safety system. This alarm will normally cause an automatic emergency stop.
1008	SD Engine Speed Camshaft	Yellow (4)	Sensor defect on camshaft. ==> short circuit or cable breakage ==> Check sensor and cable (B1), if necessary replace it. After Engine restart follows curing of failure.
1010	SD Engine Speed Crankshaft	Yellow (4)	Sensor defect on crankshaft. ==> short circuit or cable breakage ==> Check sensor and cable (B13), if necessary replace it. After Engine restart follows curing of failure.
1018	SD ETC 1 Speed	Yellow (4)	Speed-sensor of basic charger defect. ==> short circuit or cable breakage ==> Check sensor and cable (B44.1), if necessary replace it.
1020	HIHI ETC 1 Overspeed	Red (2)	Speed of basic charger too high (limit 2).
1022	HI ETC 1 Speed	Yellow (3)	Speed of basic charger too high (limit 1).
1024	SD ETC 2 Speed	Yellow (4)	Speed-sensor of switching charger defect. ==> short circuit or cable breakage ==> Check sensor and cable (B44.2), if necessary replace it.
1026	HIHI ETC 2 Overspeed	Red (2)	Speed of 1st switchable charger too high (limit 2).
1028	HI ETC 2 Speed	Yellow (3)	Speed of 1st switchable charger too high (limit 1).
1030	SD ETC 3 Speed	Yellow (4)	Speed-sensor of switching charger defect. ==> short circuit or cable breakage ==> Check sensor and cable (B44.2), if necessary replace it.
1032	HIHI ETC 3 Overspeed	Red (2)	Speed of 2nd switchable charger too high (limit 2).
1034	HI ETC 3 Speed	Yellow (3)	Speed of 2nd switchable charger too high (limit 1).
1036	SD ETC 4 Speed	Yellow (4)	Speed-sensor of switching charger defect. ==> short circuit or cable breakage ==> Check sensor and cable (B44.4), if necessary replace it.
1038	HIHI ETC 4 Overspeed	Red (2)	Speed of 3rd switchable charger too high (limit 2).
1040	HI ETC 4 Speed	Yellow (3)	Speed of 3rd switchable charger too high (limit 1).
1048	HI ETC Speed Difference	Yellow (4)	Speed deviation between basic turbo charger and one of the switchable chargers.
1110	LO Starter Supply Voltage	Yellow (4)	The starter supply voltage is below the minimum limit. The starter supply voltage is not monitored during the start procedure.
1112	SD ECU Power Supply Voltage	Yellow (4)	Internal ECU error. ==> electronic defect.
1114	HIHI ECU Power Supply Voltage	Red (2)	Power supply voltage too high (limit 2) ==> Check battery / generator
1116	HI ECU Power Supply Voltage	Yellow (4)	Power supply voltage too high (limit 1) ==> Check battery / generator
1118	LO ECU Power Supply Voltage	Yellow (4)	Power supply voltage too low (limit 1) ==> Check battery / generator
1120	LOLO ECU Power Supply Voltage	Red (2)	Power supply voltage too low (limit 2) ==> Check battery / generator
1122	SD U-Power Driver Unit	Yellow (4)	Sensor defect of Injector Power driver unit. ==> Internal error of ECU7. Change ECU7.
1124	HIHI U-Power Driver Unit	Yellow (3)	Power driver voltage (injectors) too high (limit 2).
1126	HI U-Power Driver Unit	Yellow (3)	Power driver voltage (injectors) too high (limit 1).
1128	LO U-Power Driver Unit	Yellow (4)	Power driver voltage (injectors) too low (limit 1).
1130	LOLO U-Power Driver Unit	Red (2)	Power driver voltage (injectors) too low (limit 2).
1132	HI U-SPU Main Power Supply	Yellow (3)	The voltage of the SPU main power supply is too high.
1134	LO U-SPU Main Power Supply	Yellow (3)	The voltage of the SPU main power supply is too low.

Register	Denomination	Alarm Priority	Alarm Description
1136	HI U-SPU Emerg.Power Supply	Yellow (3)	The voltage of the SPU emergency power supply is too high.
1138	LO U-SPU Emerg.Power Supply	Yellow (3)	The voltage of the SPU emergency power supply is too low.
1140	LO U-SPU Real Time Clock	Yellow (3)	The voltage of backup battery for time and date in the LOP is too low, logging features may have an incorrect time stamp. Replace the battery located in the SPU/LOP!
1142	AL EMU 11V	Yellow (4)	The internal 11V power supply of the engine monitoring extension unit (EMU) is out of range. This could cause improper readings of measured data and malfunctioning safety functions.
1144	AL EMU 7V	Yellow (4)	The internal -7V power supply of the engine monitoring extension unit (EMU) is out of range. This could cause improper readings of measured data and malfunctioning safety functions.
1146	AL U-SPU Main Power SupplyFail	Yellow (4)	The main power supply of the LOP/SPU failed. The system is still full operative and is feed through the emergency power supply.
1148	AL U-SPU Emerg. Pow Sup Fail	Yellow (4)	The emergency power supply of the LOP/SPU failed. The system is still full operative and is feed through the emergency power supply.
1150	SD P-Lube Oil	Yellow (4)	Lube oil pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B5), if necessary replace it.
1152	LO P-Lube Oil	Yellow (3)	Pressure of lube oil too low (limit 1).
1154	LOLO P-Lube Oil	Red (2)	Pressure of lube oil too low (limit 2).
1156	SD P-Lube Oil SaSy	Yellow (4)	The safety system sensor signal for lube oil pressure is outside the defined sensor signal range. The safety system lube oil pressure information is not available anymore for indication or monitoring. The safety functions related to this signal are not available anymore.
1158	LO P-Lube Oil SaSy	Yellow (3)	A low engine lube oil pressure was detected by the safety system. A worsening situation could trigger a safety reaction.
1160	LOLO P-Lube Oil SaSy	Red (2)	A very low engine lube oil pressure was detected by the safety system. This alarm will normally cause an automatic security shutdown. The safety reaction is alarmed with an separate alarm.
1162	TD P-Lube Oil	Yellow (4)	The pressure difference between the redundant lube oil pressure sensors is too high.
1164	SD P-Lube Oil A-Side ETCs	Yellow (4)	Lube oil pressure-sensor of ETC on engine A-Side is defect. ==> short circuit or cable breakage ==> Check sensor and cable if necessary replace it.
1166	LO P-Lube Oil A-Side ETCs	Yellow (3)	A low lube oil pressure of the engine turbo charger on the A-side was detected. A worsening situation could trigger a safety reaction. This is the warning level.
1168	LOLO P-Lube Oil A-Side ETCs	Red (2)	A very low lube oil pressure of the engine turbo charger on the A-side was detected. A worsening situation could trigger a safety reaction. This is the warning level. This alarm will normally cause an automatic security shutdown. The safety reaction is alarmed with a separate alarm.
1170	SD P-Lube Oil B-Side ETCs	Yellow (4)	Lube oil pressure-sensor of ETC on engine B-side is defect. ==> short circuit or cable breakage ==> Check sensor and cable if necessary replace it.
1172	LO P-Lube Oil B-Side ETCs	Yellow (3)	A low lube oil pressure of the engine turbo charger on the B-side was detected. A worsening situation could trigger a safety reaction. This is the warning level.
1174	LOLO P-Lube Oil B-Side ETCs	Red (2)	A very low lube oil pressure of the engine turbo charger on the B-side was detected. A worsening situation could trigger a safety reaction. This is the warning level. This alarm will normally cause an automatic security shutdown. The safety reaction is alarmed with an separate alarm.
1176	SD P-Oil after Level Pump	Yellow (4)	Sensor defect of lube oil pressure after level pump
1178	LO P-Oil after Level Pump	Yellow (3)	Low Limit of lube oil pressure after level pump
1180	SD P-Oil bef.High Press.Pump A	Yellow (4)	Sensor defect of lube oil pressure before high pressure pump in A
1182	LO P-Oil bef.High Press.Pump A	Yellow (3)	Low Limit of lube oil pressure before high pressure pump A
1184	SD P-Oil bef.High Press.Pump B	Yellow (4)	Sensor defect of lube oil pressure before high pressure pump in B
1186	LO P-Oil bef.High Press.Pump B	Yellow (3)	Low Limit of lube oil pressure before high pressure pump B
1204	SD P-Lube Oil before Filter	Yellow (4)	Pressure sensor for lube oil before filter defect. ==> short circuit or cable breakage ==> Check sensor and cable (B5.3), if necessary replace it.
1206	SD P-Oil Filter Difference	Yellow (4)	Sensor for differential pressure of lube oil defect. ==> short circuit or cable breakage ==> Check sensor and cable (B5), if necessary replace it.
1208	HIHI P-Oil Filter Difference	Red (2)	Differential pressure of oil filter too high (limit 2).
1210	HI P-Oil Filter Difference	Yellow (3)	Differential pressure of oil filter too high (limit 1).
1226	SD P-Oil Refill Pump	Yellow (4)	The pressure sensor signal for the oil refill pump pressure monitoring is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured (safety) functions related to this signal are not available anymore.
1228	LO P-Oil Refill Pump	Yellow (3)	The oil pressure after the oil refill pump which is pumping oil out and back into the oil pan was low for a certain period (several minutes). As long as the pump is able to pump oil out and back into the oil pan, there is enough oil in the engine. If the oil level is below the end of the "pump snorkel" the pump will not produce pressure within a certain time and this alarm will be triggered.
1232	SD P-Coolant	Yellow (4)	Coolant pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B16), if necessary replace it.
1234	LO P-Coolant	Yellow (3)	Coolant pressure too low (limit 1) ==> check cooling cycle.
1236	LOLO P-Coolant	Red (2)	Coolant pressure too low (limit 2) ==> Engine stop or limitation of the injection quantity ==> check cooling cycle
1242	SD P-Raw Water	Yellow (4)	The pressure sensor signal for the raw water pressure monitoring is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured (safety) functions related to this signal are not available anymore.
1244	LO P-Raw Water	Yellow (3)	The raw water pressure after the pump is too low
1246	SD P-Fuel after Filter	Yellow (4)	Fuel pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B34), if necessary replace it.
1248	LO P-Fuel	Yellow (3)	Fuel supply pressure too low (limit 1) ==> Check filter, fuel on low-pressure side

Register	Denomination	Alarm Priority	Alarm Description
1250	LOLO P-Fuel	Red (2)	Fuel supply pressure too low (limit 2) ==> Check filter (low-pressure side)
1252	SD P-Fuel before Filter	Yellow (4)	Fuel pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B5.3), if necessary replace it.
1254	SD Fuel Prefilter 1	Yellow (4)	The self test function of the fuel prefilter sensor 1 detected that the sensor does not respond as specified. The sensor might be defect or the wiring is defective. The SD alarm is reset if a successful self test was carried out. A self test is typically performed every 15min (the exact time might vary).
1256	SD Fuel Prefilter 2	Yellow (4)	The self test function of the fuel prefilter sensor 2 detected that the sensor does not respond as specified. The sensor might be defect or the wiring is defective. The SD alarm is reset if a successful self test was carried out. A self test is typically performed every 15min (the exact time might vary).
1258	HI P-Fuel Int. Filter Diff.	Yellow (3)	Fuel Prefilter Difference to high
1260	SD P-Fuel Filter Difference	Yellow (4)	Sensor for differential pressure of fuel defect. ==> only if SD-Alarms Fuel b. Filter or Fuel after Filter
1262	HI P-Fuel Filter Difference	Yellow (3)	Differential pressure of fuel filter too high (limit 1).
1264	SD P-Fuel bef. Low Press. Pump	Yellow (4)	Sensor defect of fuel pressure before low pressure pump
1266	LO P-Fuel bef. Low Press. Pump	Yellow (3)	Low Limit of fuel pressure before low pressure pump
1268	SD P-Fuel Common Rail	Yellow (4)	Rail pressure-sensor defect. ==> High-pressure controller emergency operation ==> short circuit or cable breakage ==> Check sensor and cable (B48), if necessary replace it.
1270	HI P-Fuel Common Rail	Yellow (3)	Rail pressure > set point value => DBR reduction, shift of start of injection delayed (==> interphase transformer sticks or connections of the interphase transformer)
1272	LOLO P-Fuel Common Rail	Red (2)	Rail pressure < setpoint value => DBR reduction (==> interphase transformer defective or leakage in high-pressure system)
1274	LO P-Fuel Common Rail	Yellow (3)	Rail pressure < set point value => DBR reduction (==> interphase transformer defective or leakage in high-pressure system)
1276	SD P-Fuel Common Rail B	Yellow (4)	Rail pressure-sensor defect. ==> High-pressure controller emergency operation ==> short circuit or cable breakage ==> Check sensor and cable (B48), if necessary replace it.
1278	HI P-Fuel Common Rail B	Yellow (3)	High Limit of fuel pressure in rail B
1280	LOLO P-Fuel Common Rail B	Red (2)	Rail pressure on Rail B < setpoint value => DBR reduction (==> interphase transformer defective or leakage in high-pressure system)
1282	LO P-Fuel Common Rail B	Yellow (3)	Rail pressure < setpoint value => DBR reduction (==> interphase transformer defective or leakage in high-pressure system)
1296	SD P-Fuel Return Path	Yellow (4)	Sensor defect of fuel pressure in return path
1298	HI P-Fuel Return Path	Yellow (3)	High Limit of fuel pressure in return path
1300	SD P-Crankcase	Yellow (4)	Crankcase pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B50), if necessary replace it.
1302	HIHI P-Crankcase	Red (2)	Crankcase pressure too high (limit 2).
1304	HI P-Crankcase	Yellow (3)	Crankcase pressure too high (limit 1).
1306	LO P-Intake Air After FilterB	Yellow (3)	Low Limit of intake air pressure B reached
1308	SD P-Charge Air A	Yellow (4)	Charge air pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B10), if necessary replace it.
1310	HIHI P-Charge Air A	Red (2)	Pressure of charge air too high (limit 2).
1312	HI P-Charge Air A	Yellow (3)	Pressure of charge air too high (limit 1).
1314	SD P-Charge Air B	Yellow (4)	Charge Air Pressur B-Side defect. ==> short circuit or cable breakage ==> Check sensor and cable (B10.11), if necessary replace it. After Engine restart follows curing of failure.
1316	HIHI P-Charge Air B	Red (2)	Highest Limit of charge air pressure in B reached
1318	HI P-Charge Air B	Yellow (3)	High Limit of charge air pressure in B reached
1324	SD P-Start Air	Yellow (4)	The pressure sensor signal for the start air pressure monitoring is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured (safety) functions related to this signal (start interlock if start air pressure is too low) are not available anymore.
1326	LO P-Start Air	Yellow (4)	The pressure for the air starter is too low.
1328	LOLO P-Start Air	Yellow (3)	The pressure for the air starter is very low and a start is interlocked. The limit for that alarm should be set in a way that air for at least one start is remaining. This is to prevent emptying the air supply by (automatic) repeated starts. The start interlock can be disabled by activating "Override"
1350	HI P-Start Air	Yellow (3)	The pressure for the air starter is too high. A high air pressure can damage the air starter.
1352	HIHI P-Start Air	Red (2)	The pressure for the air starter is too high for the air starter and a start is interlocked to protect the air starter against damange. The start interlock can be disabled by activating "Override"
1386	SD T-Lube Oil	Yellow (4)	Lube oil temperature-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B7), if necessary replace it.
1388	HIHI T-Lube Oil	Red (2)	Lube oil temperature to high (limit 2).
1390	HI T-Lube Oil	Yellow (3)	Lube oil temperature too high (limit 1).
1394	SD T-Lube Oil ETCs	Yellow (4)	T-Lube Oil turbo charger sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable, if necessary replace it.
1396	HIHI T-Lube Oil ETCs	Red (2)	Lube oil temperature of the basic charger is too high (above second limit).

Register	Denomination	Alarm Priority	Alarm Description
1398	HI T-Lube Oil ETCs	Yellow (3)	Lube oil temperature of the basic charger is too high (above first limit).
1418	SD T-Fuel bef.Low Press.Pump	Yellow (4)	T-Fuel before Engine sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable, if necessary replace it.
1420	HIHI T-Fuel before LP Pump	Red (2)	Highest Limit of fuel temperature before low pressure pump reached
1422	HI T-Fuel before LP Pump	Yellow (3)	Fuel temperature before the low pressure fuel pump is too high.
1424	SD T-Fuel	Yellow (4)	Fuel temperature-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B33), if necessary replace it.
1426	HIHI T-Fuel	Red (2)	Fuel temperature too high (limit 2).
1428	HI T-Fuel	Yellow (3)	Fuel temperature too high (limit 1)
1430	SD T-Fuel Common Rail B	Yellow (4)	Sensor defect of fuel temperature in Rail B
1432	HIHI T-Fuel Common Rail B	Red (2)	Highest Limit of fuel temperature in rail B reached
1434	HI T-Fuel Common Rail B	Yellow (3)	High Limit of fuel temperature in rail B reached
1436	SD T-Coolant	Yellow (4)	Coolant temperature-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B6), if necessary replace it.
1438	HIHI T-Coolant	Red (2)	Coolant temperature too high (limit 2) ==> Check circulation of coolant
1440	HI T-Coolant	Yellow (3)	Coolant temperature too high (limit 1) ==> Check circulation of coolant
1442	SD T-Coolant SaSy	Yellow (4)	The safety system sensor signal for coolant temperature is outside the defined sensor signal range. The safety system coolant temperature information is not available anymore for indication or monitoring. The safety functions related to this signal are not available anymore.
1444	HIHI T-Coolant SaSy	Red (2)	A very high engine coolant temperature was detected by the safety system. This alarm will normally cause an automatic safety function. The safety reaction is alarmed with an separate alarm.
1446	HI T-Coolant SaSy	Yellow (4)	A high engine coolant temperature was detected by the safety system. A worsening situation could trigger a safety reaction.
1448	TD T-Coolant	Yellow (4)	The temperature difference between the redundant coolant temperature sensors is too high.
1450	SD T-Raw Water after Pump	Yellow (4)	Sensor defect of raw water temperature after pump
1452	HI T-Raw Water after Pump	Yellow (3)	High Limit of raw water temperature after pump reached
1454	SD T-Intake Air	Yellow (4)	Intake air temperature-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B3), if necessary replace it.
1456	HIHI T-Intake Air	Red (2)	Highest Limit of intake air temperature reached
1458	HI T-Intake Air	Yellow (3)	Intake Air temperature too high (limit 1).
1460	SD P-Intake Air After FilterB	Yellow (4)	Intake air pressure sensor and cable , if necessary replace it.
1464	SD T-Charge Air	Yellow (4)	Charge air temperature-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B9), if necessary replace it.
1466	HIHI T-Charge Air	Red (2)	Charge air temperature too high (limit 2) ==> Check Intercooler
1468	HI T-Charge Air	Yellow (3)	Charge air temperature too high (limit 1) ==> Check Intercooler
1470	SD T-Charge Air B	Yellow (4)	Charge Air B temperature-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B3), if necessary replace it.
1472	HIHI T-Charge Air B	Red (2)	Highest Limit of charge air temperature in B reached
1474	HI T-Charge Air B	Yellow (3)	High Limit of charge air temperature in B reached
1476	SD T-Charge Air Seq Ctrl Valve	Yellow (4)	Temperature-sensor of recirculated charge air defect. ==> short circuit or cable breakage ==> Check sensor and cable (B49), if necessary replace it.
1478	HIHI T-Charge Air Seq Ctrl Val	Red (2)	Umblasen' temperature too high (limit 2).
1480	HI T-Charge Air Seq Ctrl Valve	Yellow (3)	Umblasen' temperature too high (limit 1).
1482	SD T-Exhaust Combined A	Yellow (4)	Exhaust gas temperature-sensor on A-side defect. ==> short circuit or cable breakage ==> Check sensor and cable (B4.21), if necessary replace it.
1484	HIHI T-Exhaust Combined A	Red (2)	Exhaust gas temperature (A-side) too high (limit 2).
1486	HI T-Exhaust Combined A	Yellow (3)	Exhaust gas temperature (A-side) too high (limit 1).
1488	SD T-Exhaust Combined B	Yellow (4)	Exhaust gas temperature-sensor on B-side defect. ==> short circuit or cable breakage ==> Check sensor and cable (B4.22), if necessary replace it.
1490	HIHI T-Exhaust Combined B	Red (2)	Exhaust gas temperature (B-side) too high (limit 2).
1492	HI T-Exhaust Combined B	Yellow (3)	Exhaust gas temperature (B-side) too high (limit 1).
1494	SD T-Exhaust Combined C	Yellow (4)	Exhaust gas temperature-sensor on C-side defect. ==> short circuit or cable breakage ==> Check sensor and cable (B4.23), if necessary replace it. After Engine restart follows curing of failure.
1496	HIHI T-Exhaust Combined C	Red (2)	Exhaust gas temperature (C-side) too high (limit 2).

Aenderungsbeschreibung/Description of Revision	Kommt vor/Frequency
war kZ	

Buchst./Rev.Ltr.	Erh-Antrags-Nr. / Promotion request No.	Aenderungs-Nr./Revision Notice No.	Bearbeitungsstatus/Lifecycle
-.3	PR085802		Freigegeben

Register	Denomination	Alarm Priority	Alarm Description
1498	HI T-Exhaust Combined C	Yellow (3)	Exhaust gas temperature (C-side) too high (limit 1).
1500	SD T-Exhaust Combined D	Yellow (4)	Exhaust gas temperature-sensor on D-side defect. ==> short circuit or cable breakage ==> Check sensor and cable (B4.24), if necessary replace it. After Engine restart follows curing of failure.
1502	HIHI T-Exhaust Combined D	Red (2)	Exhaust gas temperature (D-side) too high (limit 2).
1504	HI T-Exhaust Combined D	Yellow (3)	Exhaust gas temperature (D-side) too high (limit 1).
1506	SD T-Exhaust A1	Yellow (4)	The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore.
1508	HI T-Exhaust A1	Yellow (3)	The temperature of cylinder A1 compared to the average exhaust temperature is too high.
1510	LO T-Exhaust A1	Yellow (3)	The temperature of cylinder A1 compared to the average exhaust temperature is too low.
1512	SD T-Exhaust A2	Yellow (4)	The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore.
1514	HI T-Exhaust A2	Yellow (3)	The temperature of cylinder A2 compared to the average exhaust temperature is too high.
1516	LO T-Exhaust A2	Yellow (3)	The temperature of cylinder A2 compared to the average exhaust temperature is too low.
1518	SD T-Exhaust A3	Yellow (4)	The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore.
1520	HI T-Exhaust A3	Yellow (3)	The temperature of cylinder A3 compared to the average exhaust temperature is too high.
1522	LO T-Exhaust A3	Yellow (3)	The temperature of cylinder A3 compared to the average exhaust temperature is too low.
1524	SD T-Exhaust A4	Yellow (4)	The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore.
1526	HI T-Exhaust A4	Yellow (3)	The temperature of cylinder A4 compared to the average exhaust temperature is too high.
1528	LO T-Exhaust A4	Yellow (3)	The temperature of cylinder A4 compared to the average exhaust temperature is too low.
1530	SD T-Exhaust A5	Yellow (4)	The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore.
1532	HI T-Exhaust A5	Yellow (3)	The temperature of cylinder A5 compared to the average exhaust temperature is too high.
1534	LO T-Exhaust A5	Yellow (3)	The temperature of cylinder A5 compared to the average exhaust temperature is too low.
1536	SD T-Exhaust A6	Yellow (4)	The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore.
1538	HI T-Exhaust A6	Yellow (3)	The temperature of cylinder A6 compared to the average exhaust temperature is too high.
1540	LO T-Exhaust A6	Yellow (3)	The temperature of cylinder A6 compared to the average exhaust temperature is too low.
1542	SD T-Exhaust A7	Yellow (4)	The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore.
1544	HI T-Exhaust A7	Yellow (3)	The temperature of cylinder A7 compared to the average exhaust temperature is too high.
1546	LO T-Exhaust A7	Yellow (3)	The temperature of cylinder A7 compared to the average exhaust temperature is too low.
1548	SD T-Exhaust A8	Yellow (4)	The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore.
1550	HI T-Exhaust A8	Yellow (3)	The temperature of cylinder A8 compared to the average exhaust temperature is too high.
1552	LO T-Exhaust A8	Yellow (3)	The temperature of cylinder A8 compared to the average exhaust temperature is too low.
1554	SD T-Exhaust A9	Yellow (4)	The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore.
1556	HI T-Exhaust A9	Yellow (3)	The temperature of cylinder A9 compared to the average exhaust temperature is too high.
1558	LO T-Exhaust A9	Yellow (3)	The temperature of cylinder A9 compared to the average exhaust temperature is too low.
1560	SD T-Exhaust A10	Yellow (4)	The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore.
1562	HI T-Exhaust A10	Yellow (3)	The temperature of cylinder A10 compared to the average exhaust temperature is too high.
1564	LO T-Exhaust A10	Yellow (3)	The temperature of cylinder A10 compared to the average exhaust temperature is too low.
1566	SD T-Exhaust B1	Yellow (4)	The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore.
1568	HI T-Exhaust B1	Yellow (3)	The temperature of cylinder B1 compared to the average exhaust temperature is too high.
1570	LO T-Exhaust B1	Yellow (3)	The temperature of cylinder B1 compared to the average exhaust temperature is too low.